Euclid, Book I, Proposition I.1
To Construct an Equilateral Triangle on a Given Finite Straight Line

CGJteamLab

August 7, 2026

Euclid's first proposition in Book I is usually presented as the straightforward construction obtained from the intersection of two circles. A formal reconstruction, however, reveals two logically independent problems: the existence of a common point of the corresponding circles and the proof that the resulting point does not lie on the line determined by the given base.

The proof of non-collinearity relies on the classical idea that a proper part of a segment cannot be congruent to the entire segment. This argument is well known from modern axiomatic reconstructions of geometry, including the work of Michael Beeson and his collaborators.

The purpose of this note is not to present a new geometric proof. Rather, it is to demonstrate that, once Book Zero has been developed, this argument becomes a direct application of the previously established theory. Proposition I.1 therefore serves as the first experimental test of Book Zero as an intermediate language for synthetic proofs.

1 The Classical Construction

Let $A \neq B$. Euclid constructs two circles of radius AB: the first centered at A, the second centered at B. One of their points of intersection is denoted by C.

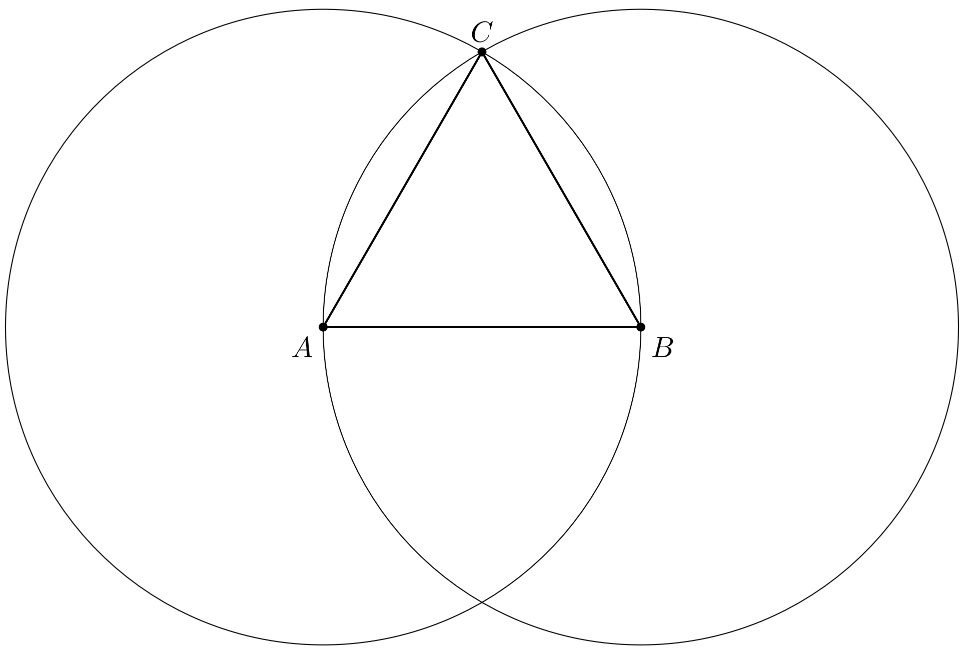


Figure 1: Euclid's classical construction. The point C is a common point of two circles of radius AB.

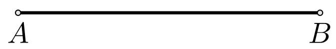
From the construction we obtain

$$AC \cong AB, \qquad BC \cong AB.$$

Formally, however, this is not yet a complete proof. One must also show that the points A, B, C are not collinear.

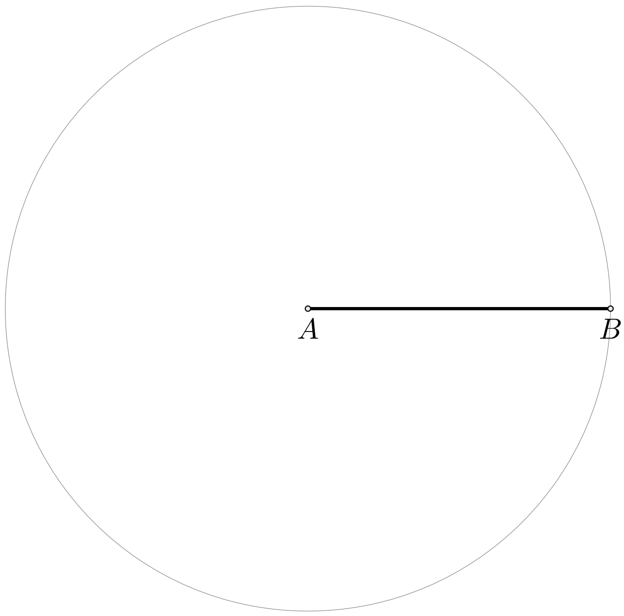
1.0.1 Step 1. Given Segment

The construction starts with a given segment AB.



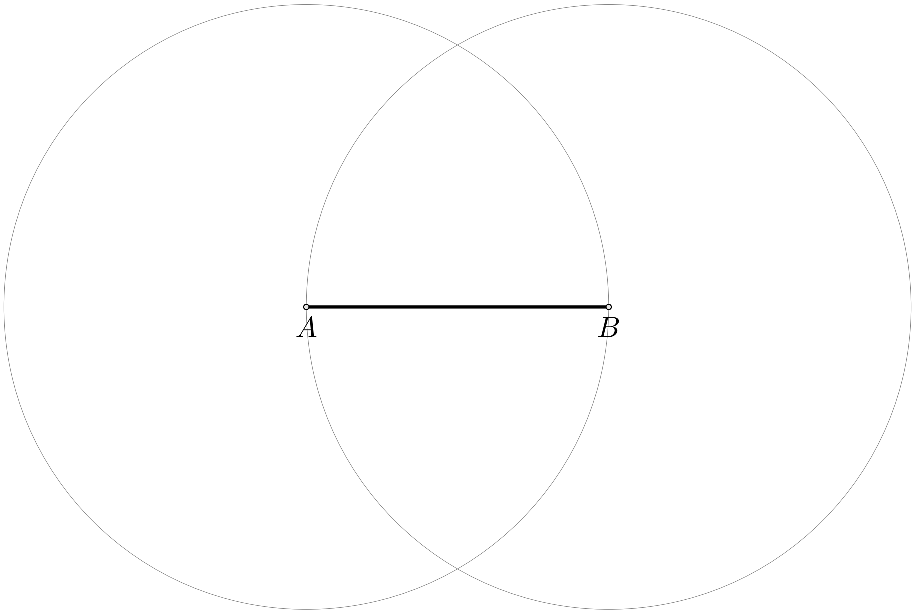
1.0.2 Step 2. Draw the Circle Centered at A

Draw the circle centered at A through B.



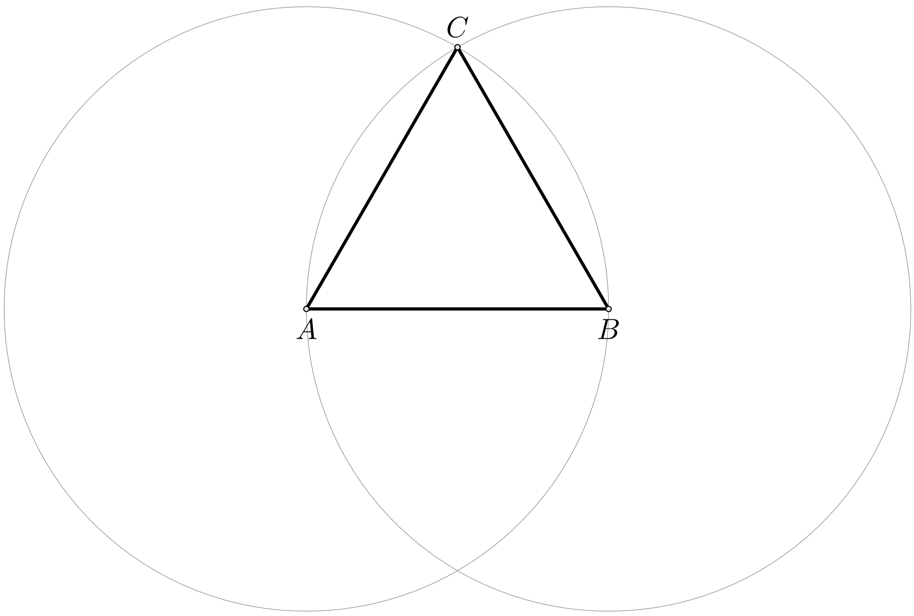
1.0.3 Step 3. Draw the Circle Centered at B

Draw the circle centered at B through A.



1.0.4 Step 4. Construct the Equilateral Triangle

Let C be one of the intersection points of the two circles. Join AC and BC to obtain the equilateral triangle.



2 Separating the Two Problems

The reconstruction reveals two logically independent stages.

1. **Existence stage.** One must obtain a point C such that

$$AC \cong AB, \qquad BC \cong AB.$$

   In the current version of the project, this stage is provided by the temporary principle `hilbert_equidistant_point_exists`. Ultimately, it is intended to be derived from an appropriate theory of continuity in Hilbert geometry.

2. **Synthetic stage.** From the congruences above one must derive

$$\neg\,\mathrm{Collinear}(A, B, C).$$

   This stage no longer requires any theory of circles.

3 Theorem

Theorem. Let $A \neq B$. Assume that there exists a point C such that

$$AC \cong AB, \qquad BC \cong AB.$$

Then the points A, B, C are not collinear.

4 Proof

First observe that $C \neq A$ and $C \neq B$. If $C = A$, then $AC \cong AB$ would imply that a null segment is congruent to the non-null segment AB. Similarly, $C \neq B$.

Now suppose, for contradiction, that the points A, B, C are collinear. Since they are pairwise distinct, the trichotomy of the order of points on a line gives exactly three possibilities:

$$A - B - C, \qquad B - A - C, \qquad A - C - B.$$

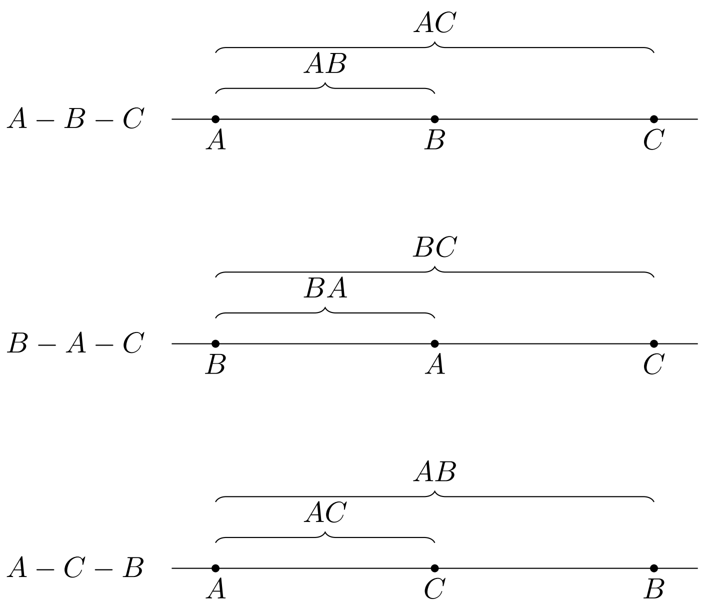


Figure 2: The three possible arrangements of pairwise distinct collinear points.

Case 1: $A - B - C$

The segment AB is a proper part of the segment AC. Book Zero contains the general theorem:

> A proper part of a segment cannot be congruent to the whole segment.

On the other hand, by assumption,

$$AC \cong AB.$$

Using symmetry of congruence, we obtain

$$AB \cong AC,$$

which is a contradiction.

Case 2: $B - A - C$

The segment BA is a proper part of the segment BC. Since

$$BC \cong AB,$$

and reversing the endpoints of a segment does not change its length, we have

$$BA \cong AB.$$

By transitivity of congruence, it follows that

$$BA \cong BC,$$

again contradicting the fact that a proper part cannot be congruent to the whole.

Case 3: $A - C - B$

The segment AC is a proper part of the segment AB. But the assumption directly gives

$$AC \cong AB.$$

This yields the third contradiction.

All possible collinear arrangements lead to a contradiction. Therefore,

$$\neg\,\mathrm{Collinear}(A, B, C).$$

$\square$

5 Architecture of the Proof

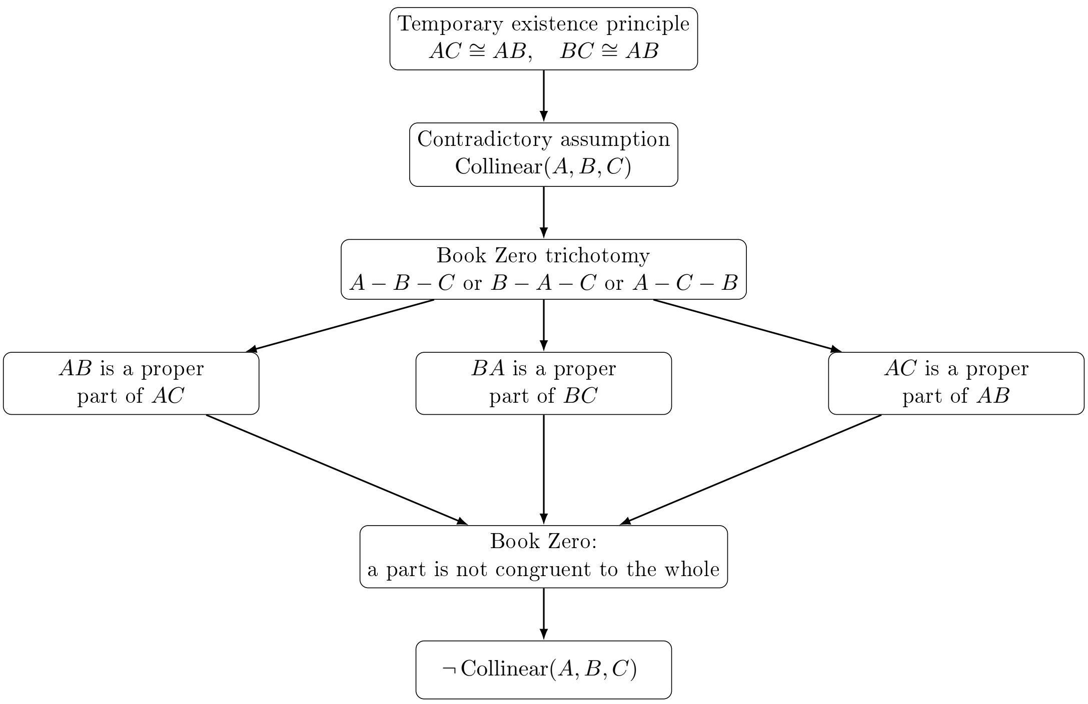


Figure 3: The structure of the proof of Proposition I.1 above the Book Zero layer.

6 Significance of the Reconstruction

The geometric result itself is not new. Constructing an equilateral triangle on a given segment is one of the classical theorems of elementary geometry.

Likewise, the idea underlying the proof of non-collinearity is not new. Modern axiomatic reconstructions, in particular the work of Michael Beeson and his collaborators, employ an analogous argument based on the theorem that a proper part of a segment cannot be congruent to the entire segment.

The novelty of the present reconstruction lies elsewhere.

The objective of the CGJteamLab project was not to discover a new proof of Proposition I.1, but to investigate whether the theory developed in Book Zero is sufficiently rich to serve as an intermediate language for synthetic proofs.

The experiment shows that, once the existence problem for an equidistant point has been isolated, the entire argument can be expressed almost exclusively in terms of the theorems previously established in Book Zero.

From the perspective of the architecture of the theory, this is more significant than Proposition I.1 itself. It demonstrates that Book Zero is not merely a collection of auxiliary lemmas, but rather forms a new layer of abstraction between Hilbert's axioms and classical synthetic reasoning.

Schematically, the situation may be represented as

$$\text{Hilbert's axioms} \longrightarrow \text{Book Zero} \longrightarrow \text{Proposition I.1.}$$

In this sense, Proposition I.1 constitutes the first experimental test confirming that Book Zero can serve as a language of synthetic proofs for the further reconstruction of Euclidean geometry.